

EARLY BREAST CANCER DETECTION USING COMPLEMENTARY MULTIMODALITY IMAGING FROM CONVENTIONAL TO ADVANCE TO ACHIEVED BETTER PROGNOSIS

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INTRODUCTION

Finding of small suspicious non palpable lesions in patient with atypical symptom often become a diagnostic challenge. Multimodality imaging is needed to help establish the diagnosis before patient undergoing invasive procedures. Correct diagnosis of small lesions ensures a good prognosis.

CASE PRESENTATION

A 64-year-old female patient presented with left shoulder pain. Breast ultrasound was performed to rule out pathology, chosen because mammography is not available in the first hospital. It revealed a small, non-palpable, irregular, suspicious lesion at the 2 o'clock, 5 cm from the left nipple, measuring 4.7 mm.

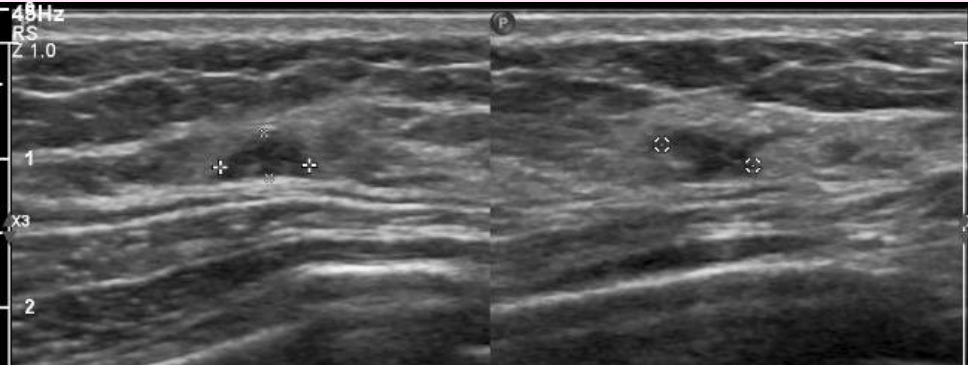


Fig 1. Initial breast US showing a small, non palpable, irregular, suspicious lesion with angular margins at the 2 o'clock position, 5 cm from the left nipple, measuring 4.7 mm.

Digital breast tomosynthesis (DBT) in the other hospital showed heterogeneous breast composition, without focal asymmetry or architectural distortion.

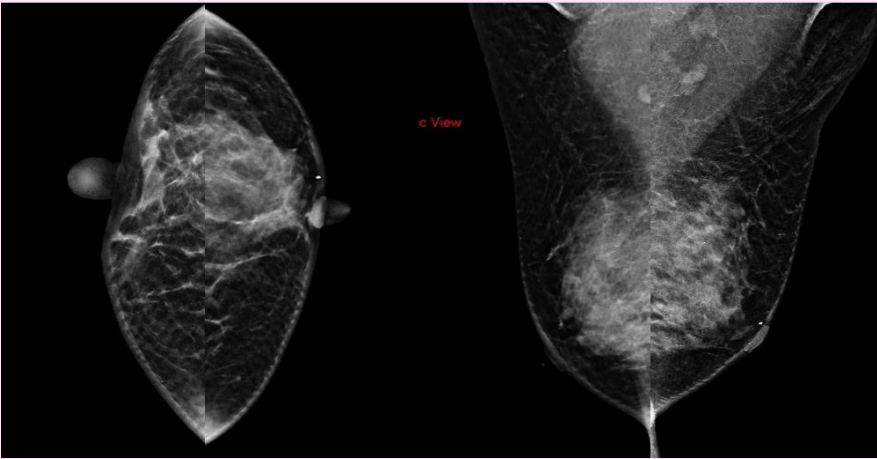


Fig 2. DBT showing a heterogeneous breast composition, without focal asymmetry or architectural distortion.

Contrast-enhanced breast MRI revealed an enhanced, irregular lesion with indistinct margins measuring 7.4 mm with type I kinetic curve. The lesion was categorized as low suspicion for malignancy (BI-RADS 4A).

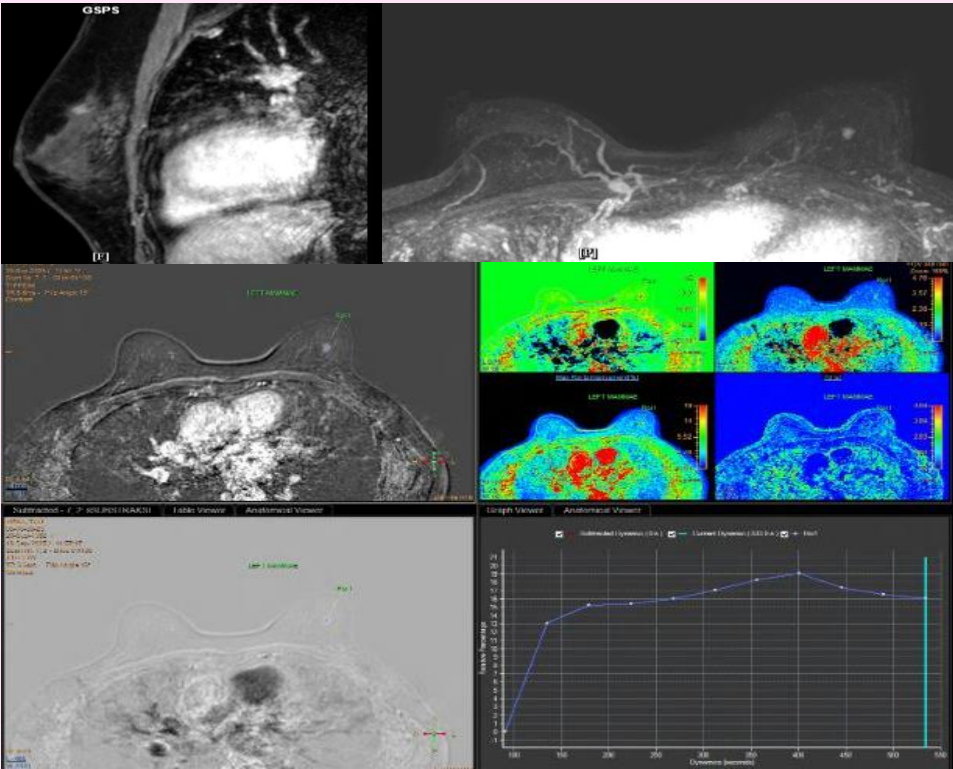


Fig 3. Breast MRI : Irregular lesion with indistinct margins with type I kinetic curve.

Patient refused biopsy or lumpectomy recommended by the oncologist. A 6-month ultrasound follow-up revealed an enlarged lesion with one new suspicious nodule. Breast-conserving surgery was performed, but the incision margins were not tumor-free, leading to a modified mastectomy.

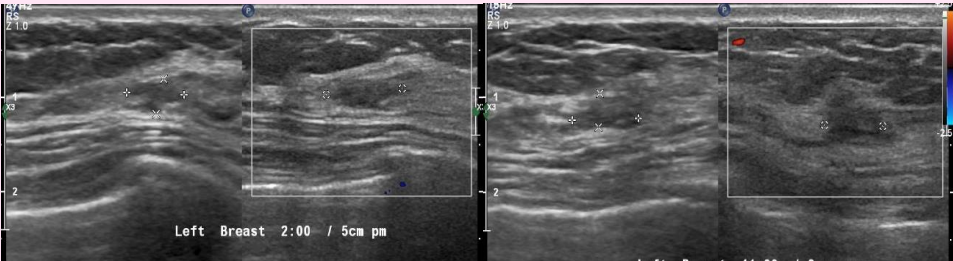


Fig 4. 6-month breast ultrasound follow-up revealed an enlarged lesion with one new suspicious nodule.

Pathology revealed a grade 3 multifocal invasive carcinoma of no special type with DCIS without axillary lymph node involvement. Post-mastectomy PET/CT scan showed no axillary lymph node spread or distant metastases.

DISCUSSION

Detecting breast carcinoma at early stage with mammography and breast ultrasound is crucial for ensuring good clinical outcome. Mammography in dense breasts may be low sensitive for detecting masses, but can assess the presence of microcalcification or architectural distortion. In early breast cancer with small lesions, if mammography and ultrasound are inconclusive, MRI is required to help confirm the diagnosis before invasive procedures. Post-contrast enhancement on breast MRI can detect lesions not visible on mammography.

CONCLUSION

Multimodality imaging (ultrasound, mammography/DBT, MRI breast) is needed for the management of patients with early breast cancer to achieve a good prognostic outcome. Breast ultrasound plays a role in the detailed morphological assessment of small non palpable lesions suggestive of malignancy, while advanced imaging such as MRI is necessary for imaging of occult early-stage breast cancer.

REFERENCES

1.

Fischer, U. Breast MRI – The champion in the millimeter league MIO breast MRI – The method of choice in women with dense breasts. European Journal of Radiology. 2023; 167.

2.

Mann RM, Cho N, Moy L. Breast MRI : State of the Art. Radiology 2019; 292:520–536

3.

Korraphong P, Trintatn O, Tangcharoensathien W, Angsusingha T, Chuthapisith S. Ultrasonographic Characteristics of Mammographically Occult Small Breast Cancer. Journal of Breast Cancer 2012 September;15(3): 344-349

4.

Wang Y, Liu D, Xu P, Fu L, Ding H. Tackling “Small” Breast Cancer in Ultrasonography: What Are They and Why Does Screening Them Help?. Research Square, 2024.